INTRODUCTION TO GENERAL CHEMISTRY 1



WHAT ARE YOU MADE OF?

YOUR NAME

YOUR STRENGTH(S)

YOUR WEAKNESS(ES)

YOUR MOTIVATION(S)

YOUR SHORT TERM GOAL(S)

YOUR LONG TERM GOAL(S)

Learning Objectives:





Compare and contrast various states of matter (solid, liquid, gas, plasma) in terms of shape, volume, compressibility, and particle movement.

Show appreciation for the role of matter and its properties in daily life and various fields (e.g., medicine, engineering, environmental science).

Perform comparative tests (e.g., heating different substances to observe changes in state or color) and identify whether changes are physical or chemical.





WHAT IS GENERAL CHEMISTRY?





- change in matter
- intro to all areas of chemistry
- laboratory environment
- experiments





WHY IS IT IMPORTANT?





- deeper understanding
- know the composition of drugs or medicine
- chemical properties
- sterilizing and sanitation
- diagnosis of disease
- discover new drugs
- how the drug acts







GENERAL CHEMISTRY 1



"khēmeia" = transmutation (change of form) of Earth.







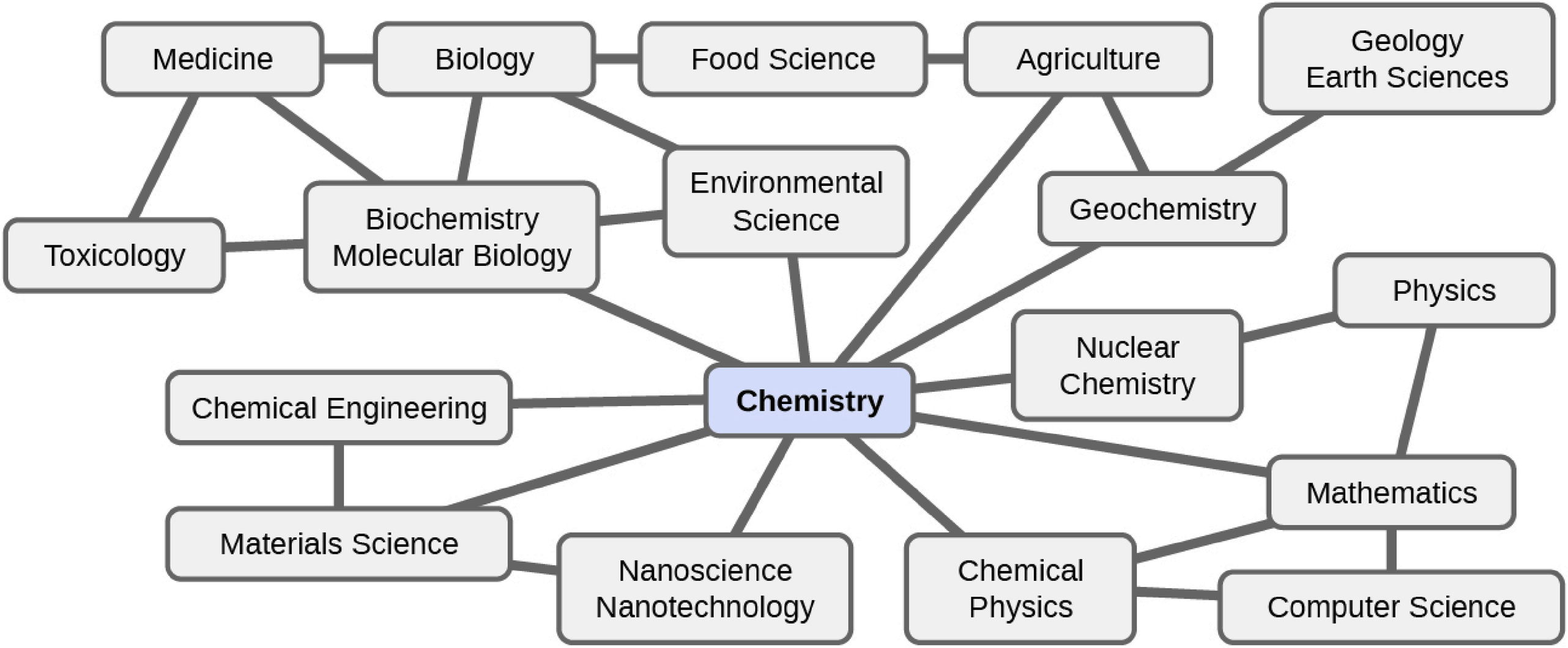
LEARNING IN ACTION



Think of at least 10 careers/jobs/fields where General Chemistry is useful and explain in what way is GenChem utilized in these fields.






Medicine
Biology
Food Science
Agriculture
Geology
Earth Sciences
Toxicology
Biochemistry
Molecular Biology
Environmental
Science
Geochemistry
Physics
Nuclear
Chemistry
Chemical Engineering
Chemistry
Mathematics
Materials Science
Nanoscience
Nanotechnology
Chemical
Physics
Computer Science

1. **Analytical chemistry** uses qualitative and quantitative observation to identify and measure the physical and chemical properties of substances. In a sense, all chemistry is analytical.
2. **Physical chemistry** combines chemistry with physics. Physical chemists study how matter and energy interact. Thermodynamics and quantum mechanics are two of the important branches of physical chemistry.
3. **Organic chemistry** specifically studies compounds that contain the element carbon. Carbon has many unique properties that allow it to form complex chemical bonds and very large molecules. Organic chemistry is known as the "Chemistry of Life" because all of the molecules that make up living tissue have carbon as part of their makeup.
4. **Inorganic chemistry** studies materials such as metals and gases that do not have carbon as part of their makeup.
5. **Biochemistry** is the study of chemical processes that occur within living organisms.

- To know the composition of drugs or medicines. eg. Drug contents
- To understand the chemical properties of the drug. Eg. Alkaline or acidic, water soluble or fat soluble.
- Predicting the drug interaction. Eg. Reaction of an antibiotic with an antacid given for ulcer.
- For sterilizing and sanitation. Eg. Alcohol, phenol, acids used for cleaning.
- In diagnosis of disease. Eg. Use for glucose strips for checking blood sugar.
- To regulate drug distribution. Eg. High dose of as drug which makes blood acidic can be stabilized by an recommended alkaline drug.
- Decrease the toxic effect.
- To discover new drugs with better effectiveness.
- To study the mechanism of disease.
- To know how the drug acts.



INTRODUCTION TO MATTER



- anything that takes up space and has mass
- made of molecules
- composed of atoms
- molecules are either moving or in stasis

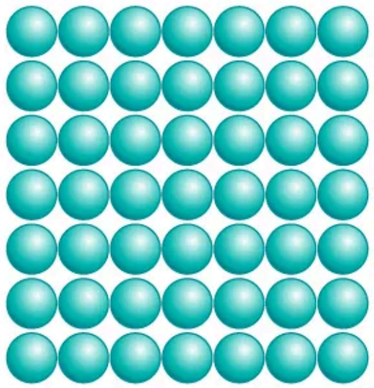




STATES OF MATTER

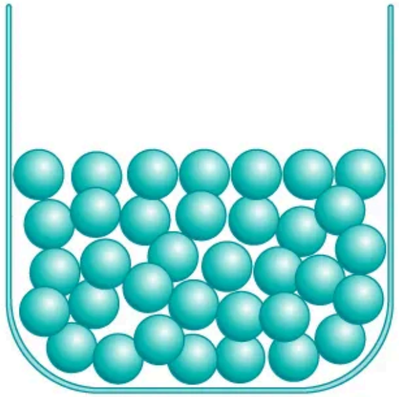
Physical states

increasing energy



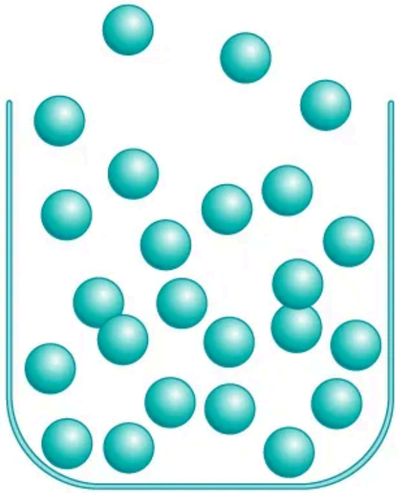
Solid

The molecules that make up a solid are arranged in regular, repeating patterns. They are held firmly in place but can vibrate within a limited area.



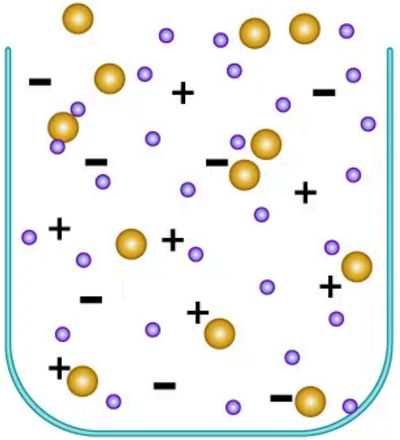
Liquid

The molecules that make up a liquid flow easily around one another. They are kept from flying apart by attractive forces between them. Liquids assume the shape of their containers.



Gas

The molecules that make up a gas fly in all directions at great speeds. They are so far apart that the attractive forces between them are insignificant.



Plasma

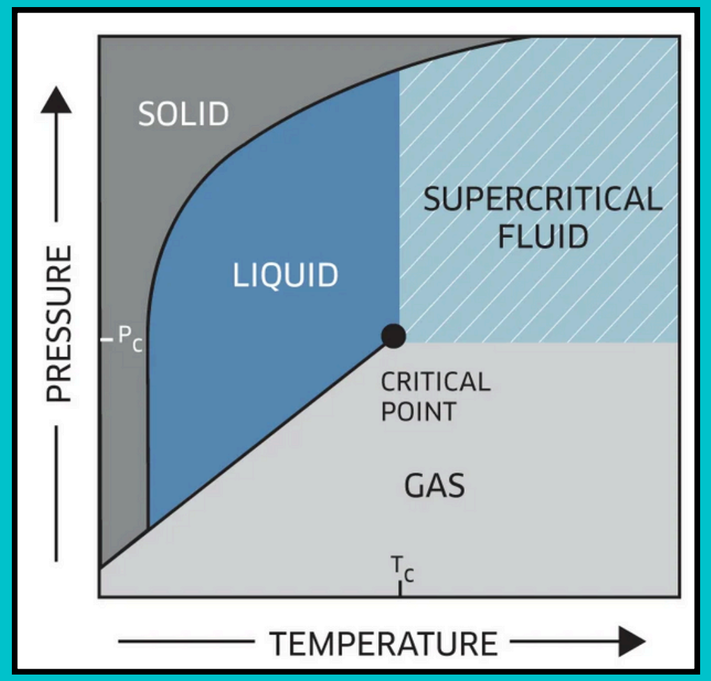
At the very high temperatures of stars, atoms lose their electrons. The mixture of electrons and nuclei that results is the plasma state of matter.

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STATES OF MATTER

Supercritical Fluids

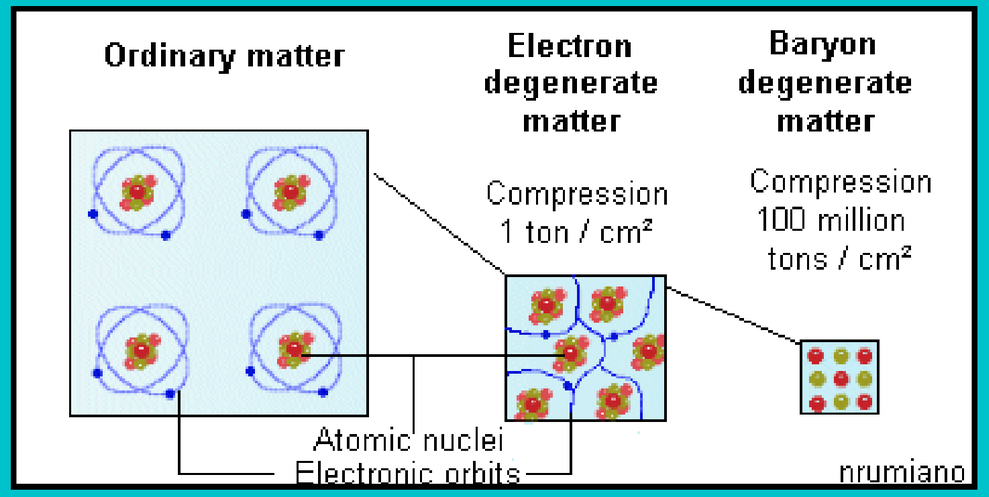
are highly compressed gases which combine properties of gases and liquids.



STATES OF MATTER

Degenerate matter

Is basically a collapsed state of matter. This happens when the atomic structure has broken down because electromagnetic forces are overcome by gravity , this explain the existence of “black holes “white dwarf starts.



GENERAL PROPERTIES OF MATTER





- Matter occupies space.
- Matter has inertia.
- Matter has mass.
- Matter is indestructible.







PROPERTIES OF MATTER



PHYSICAL PROPERTIES

(observable and measurable)

INTENSIVE PROPERTIES

do not depend on the amount of matter.
Examples: color ,odor ,luster ,malleability ,ductility .hardness etc







PROPERTIES OF MATTER

PHYSICAL PROPERTIES

(observable and measurable)



EXTENSIVE PROPERTIES

depend on the amount of matter.
Examples: mass, weight, volume and length

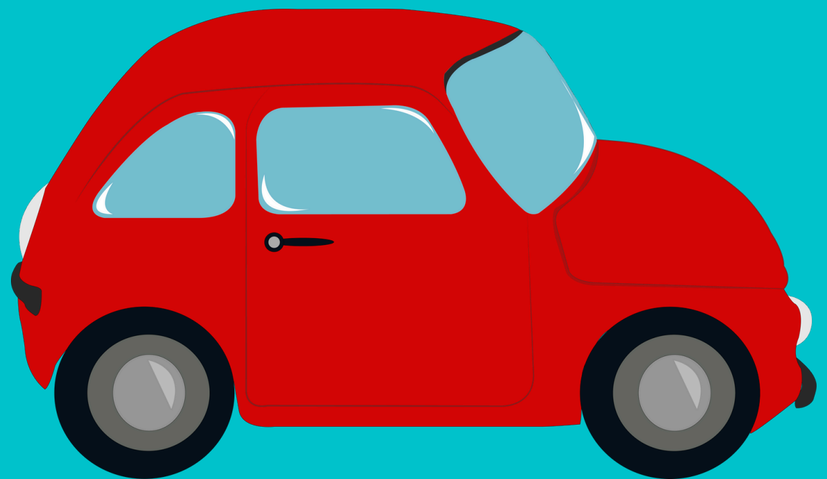




PROPERTIES OF MATTER

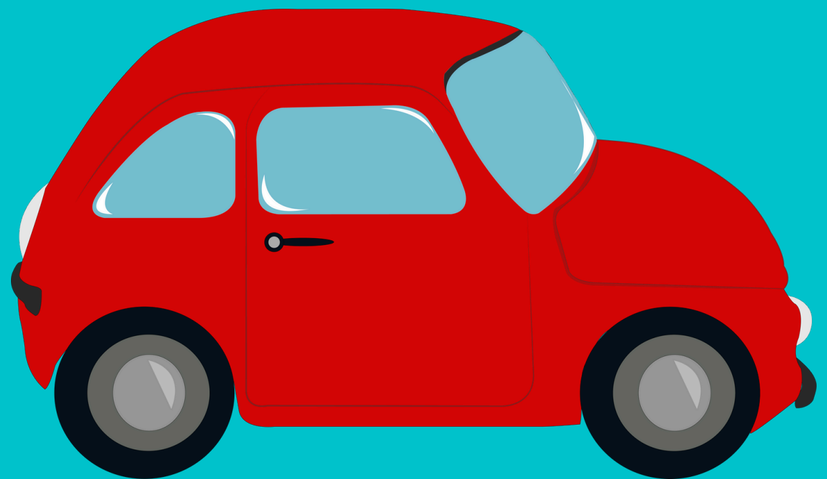


STOP THE CAR!



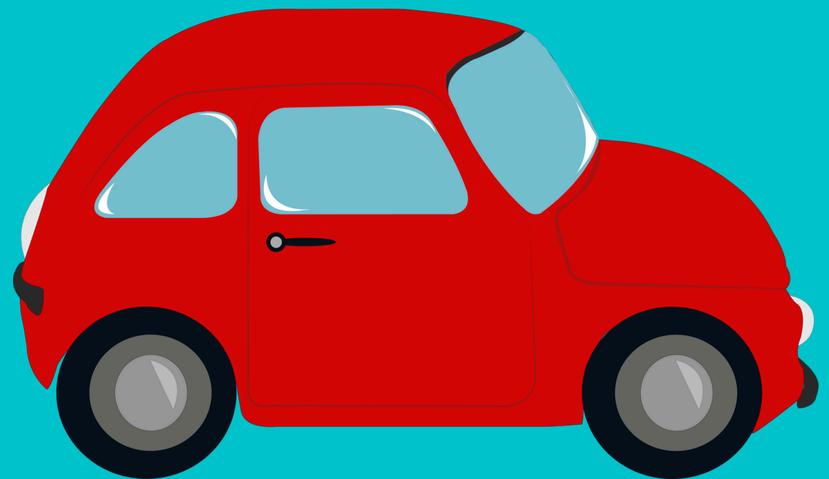


CELLPHONE BRAND



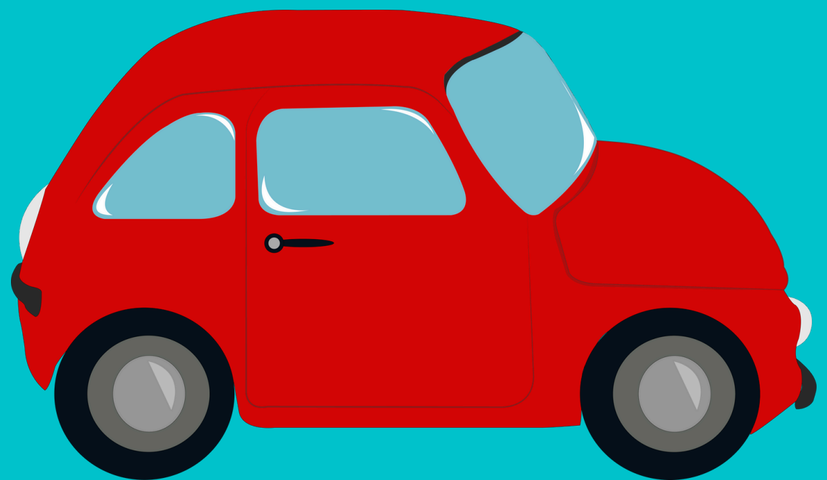


Give 3 importance of General Chemistry.



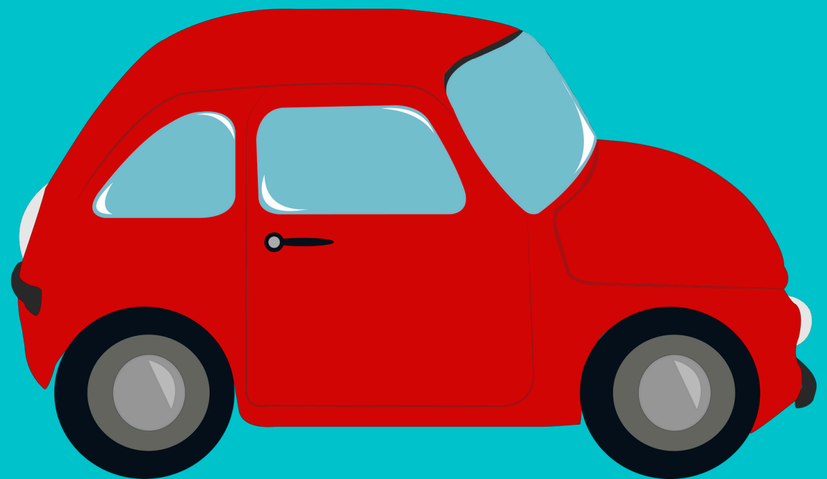


K-Pop Groups



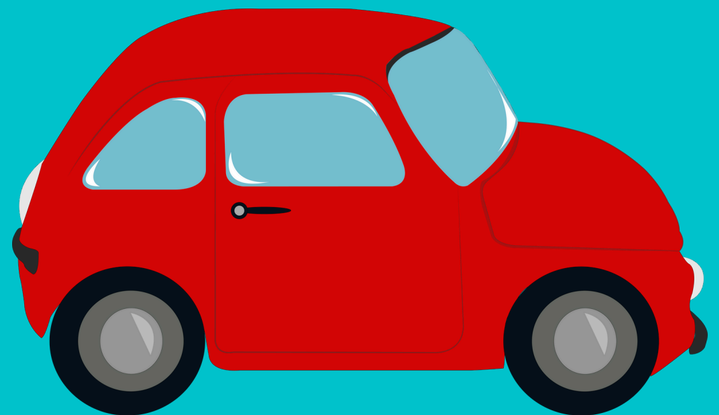


How does studying chemistry help you become a more responsible and informed citizen?



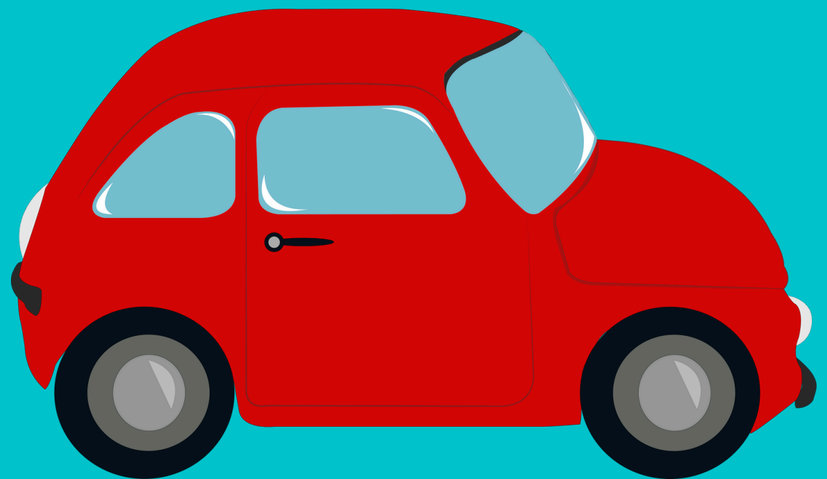


Words that start with letter “L” (Tagalog or English)



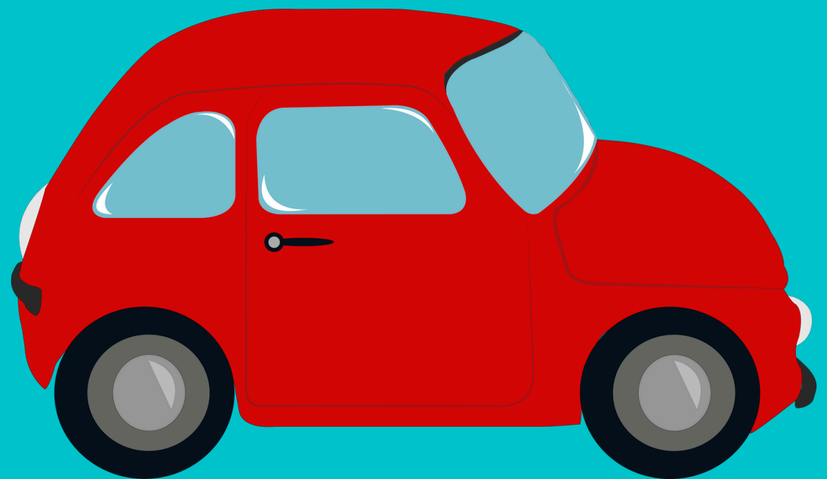


How does a pharmacist use chemistry knowledge in their profession?



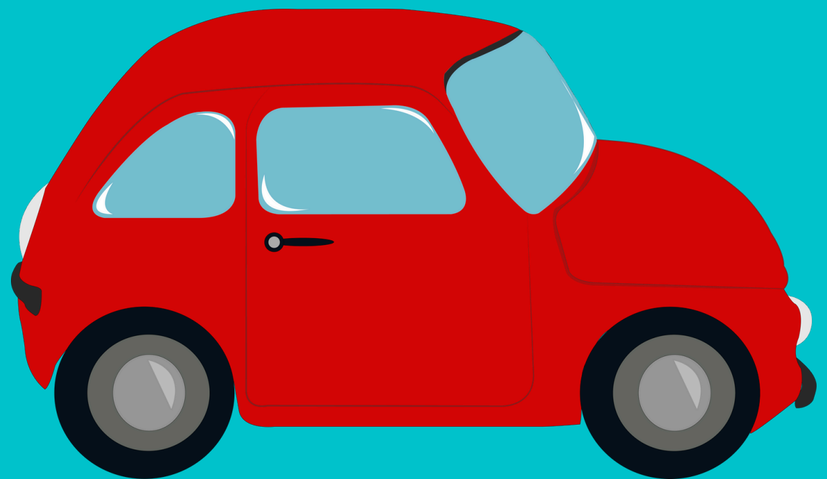


Netflix movies/TV shows



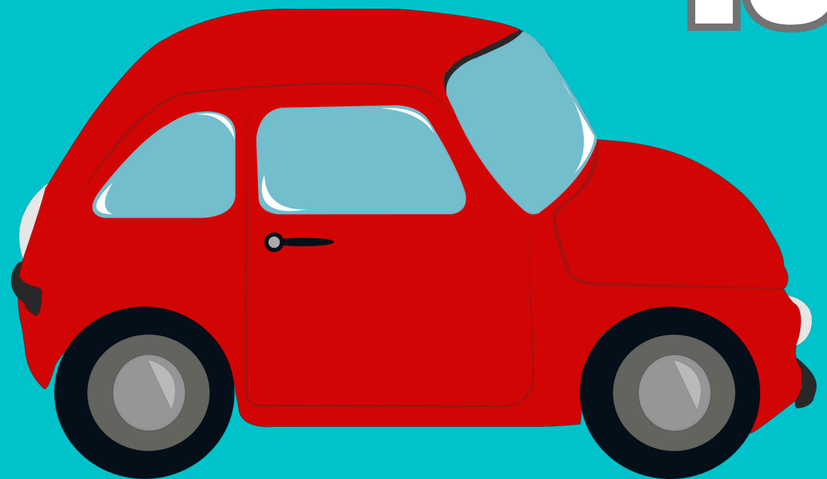


Is chemistry the most important science in modern society? Why or why not?



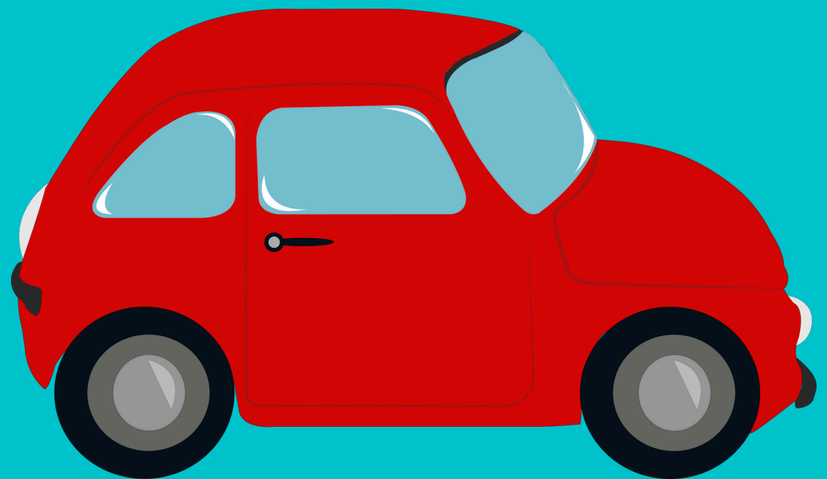


Names that starts with letter “X”





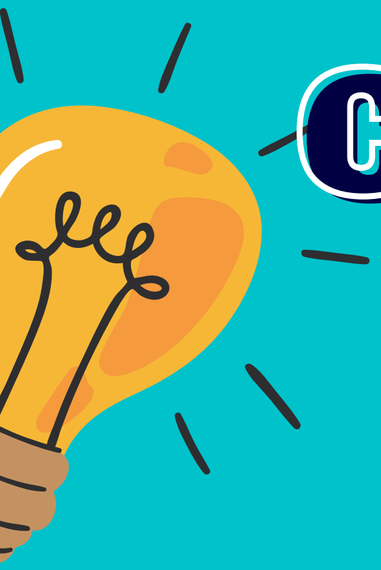
How might engineers or architects use chemistry in designing safe and sustainable structures?





CHANGES IN MATTER

Physical change





is any change in matter that involves substance going from one physical state to another.







PROPERTIES OF MATTER



PHYSICAL PROPERTIES

(observable and measurable)

INTENSIVE PROPERTIES

do not depend on the amount of matter.
Examples: color ,odor ,luster ,malleability ,ductility .hardness etc







PROPERTIES OF MATTER

PHYSICAL PROPERTIES

(observable and measurable)

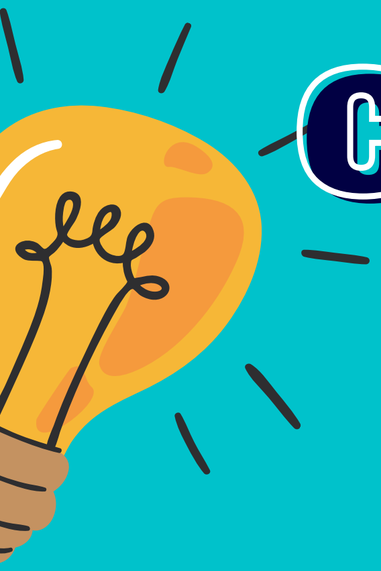


EXTENSIVE PROPERTIES

depend on the amount of matter.
Examples: mass, weight, volume and length







CHANGES IN MATTER

Physical change



Give 10 examples of Physical Change





Physical Properties



Luster



Hardness



Ductile



Malleable



Melting Point

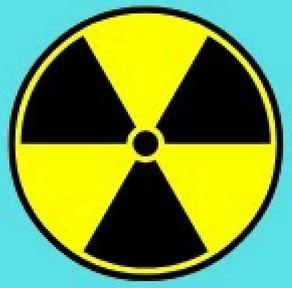


Density

Chemical Properties



Flammable



Radioactive



Corrosive

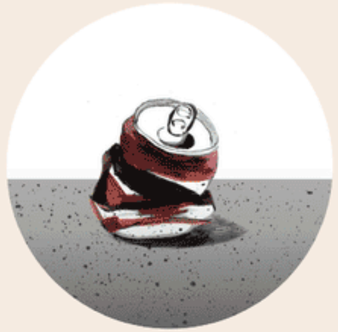


Toxic

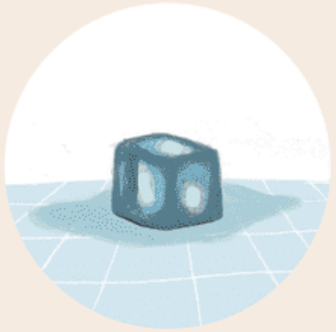


PH

Physical Changes



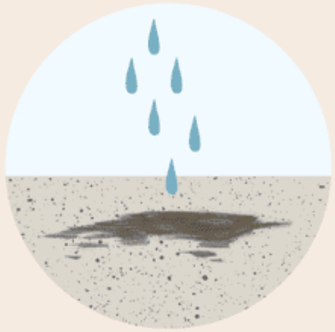
Crushing a can



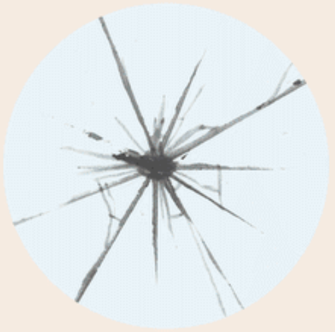
Melting an ice cube



Boiling water



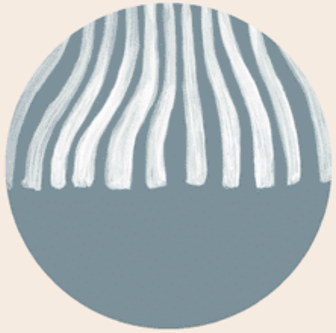
Mixing sand with water



Breaking glass



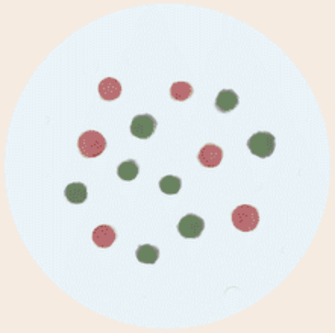
Dissolving sugar in water



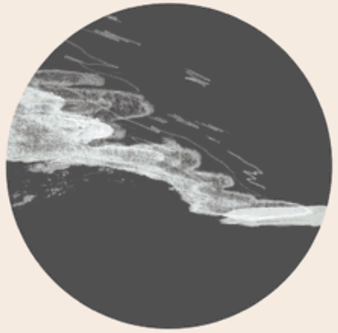
Shredding paper



Chopping wood



Mixing green and red marbles



Sublimation of dry ice

ThoughtCo.



PROPERTIES OF MATTER

CHEMICAL PROPERTIES

(observable and measurable)



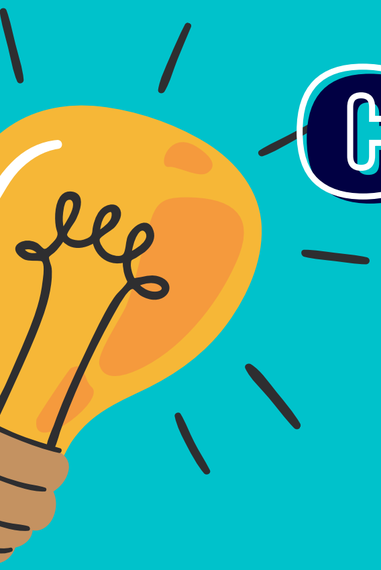
are characteristic of matter that it exhibits when it undergoes a change in composition . (Behavior of the substance in the presence with the other substance)





CHANGES IN MATTER

Chemical change





occur when a substance combines with another to form a new substance also called chemical synthesis





EVIDENCES OF CHEMICAL CHANGE





- A color change
- An odor change
- Formation of a precipitate (you mix two liquids and make a solid
- Gas is formed (bubbles)
- Changes in physical properties





CLASSES AND CLASSIFICATION



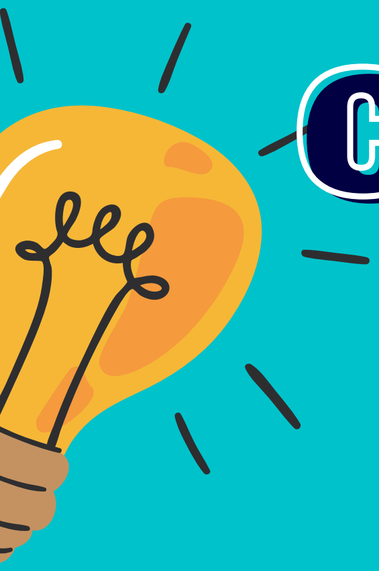


PURE SUBSTANCE

are substances that has constant composition, properties and contains only one kind of particles.







CHANGES IN MATTER

Chemical change



Give 10 examples of Chemical Change





Chemical Changes



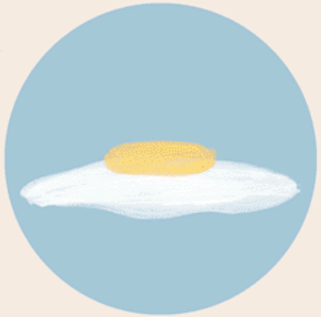
Iron Rusting



Burning Wood



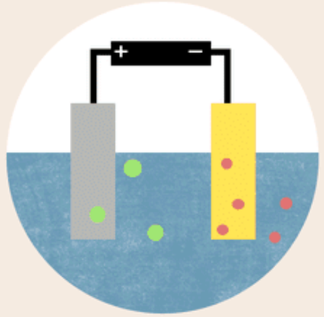
Metabolism



Cooking an Egg



Baking a Cake



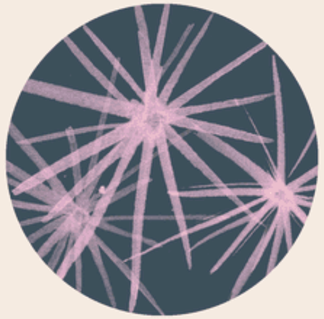
Electroplating



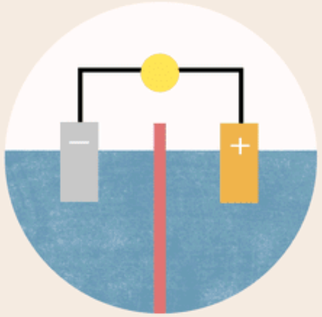
Rotting Banana



Vinegar and Baking Soda Mixture



Fireworks



Chemical Battery

ThoughtCo.

Physical Changes	Chemical Changes
• No new matter is produced. • Only physical property is changed-The visible appearance will change with out changing the composition of the matter. • No change in molecular structure. • No change occurs in the chemical composition. • Can be reversible or irreversible. • Change in temperature may or may not occur. • A matter can be physically change by boiling, melting, cutting, splitting, cracking	• One or more completely different and new matter is formed. • Changes occur both in physical and chemical properties. • Molecular structure is changed and new molecular structure are formed. • Chemical composition is changed . • Can not be reversed back to its old matter. • Change in temperature occurs in all chemical changes. • A matter can be chemically changed by burning, rusting , decomposing, fermenting

CLASSIFICATION OF MATTER

PURE SUBSTANCE

Is a type of matter that has the same composition and properties throughout. It's made of only one kind of particle.

CLASSIFICATION OF MATTER

PURE SUBSTANCE (EXAMPLES)

- Oxygen gas (O_2) – only oxygen molecules.
- Gold (Au) – only gold atoms.

CLASSIFICATION OF MATTER

PURE SUBSTANCE (EXAMPLES)





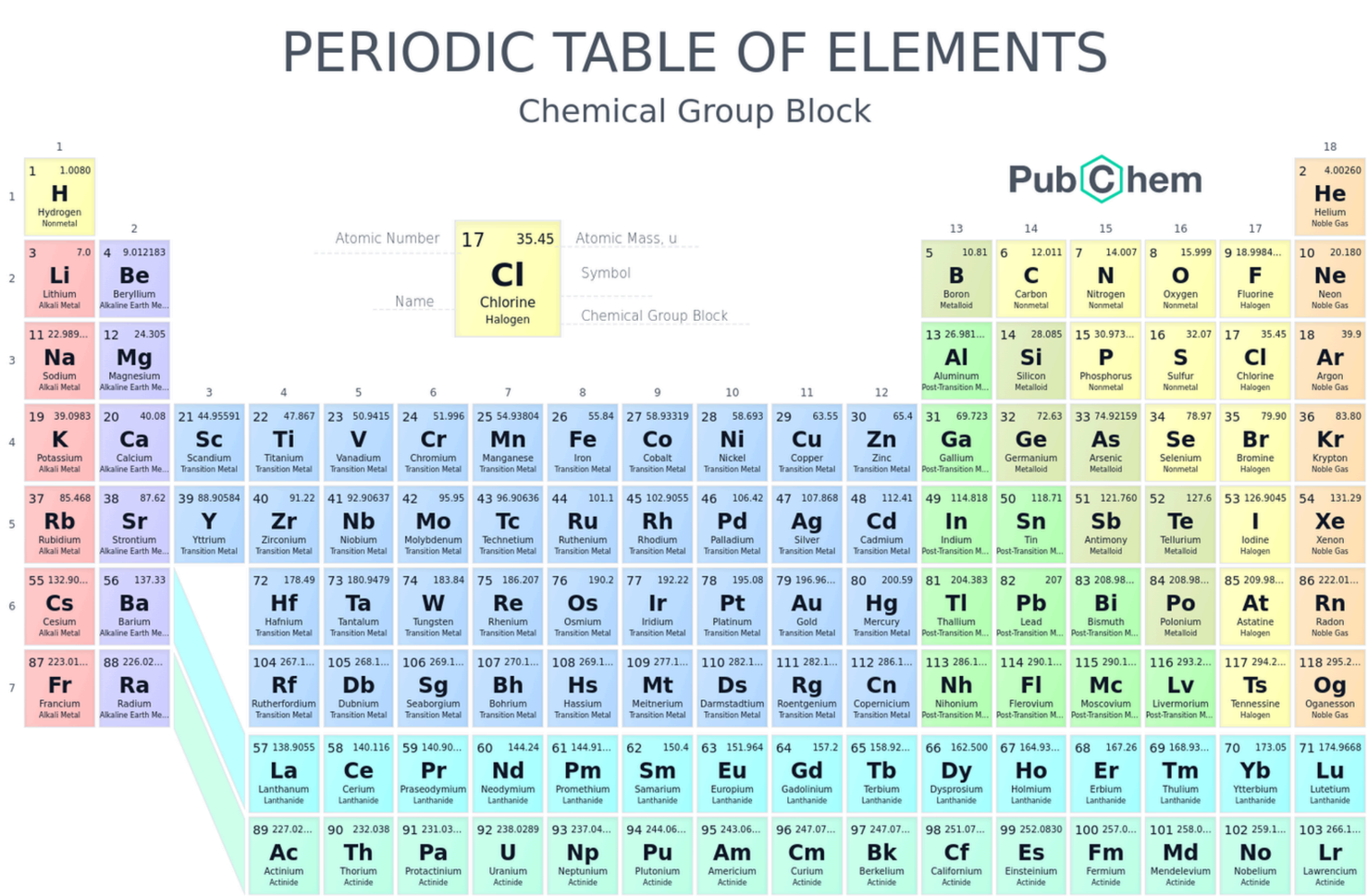
CLASSIFICATION OF MATTER

TYPES OF PURE SUBSTANCE

- **Elements - are pure substances made up of only one kind of atom. They are the basic building blocks of all matter.**

CLASSIFICATION OF MATTER

TYPES OF PURE SUBSTANCE



CLASSIFICATION OF MATTER

TYPES OF PURE SUBSTANCE

- **Elements - are pure substances made up of only one kind of atom. They are the basic building blocks of all matter.**

CLASSIFICATION OF MATTER

2 TYPES OF ELEMENTS

- **Metals - is a type of element that is usually shiny, hard, and a good conductor of heat and electricit**

CLASSIFICATION OF MATTER

TYPES OF METALS



CLASSIFICATION OF MATTER

2 TYPES OF ELEMENTS

- **Non-metals - are elements that are usually dull, brittle, and poor conductors of heat and electricity.**

CLASSIFICATION OF MATTER

TYPES OF NON-METALS

Carbon



Phosphorus



Sulfur



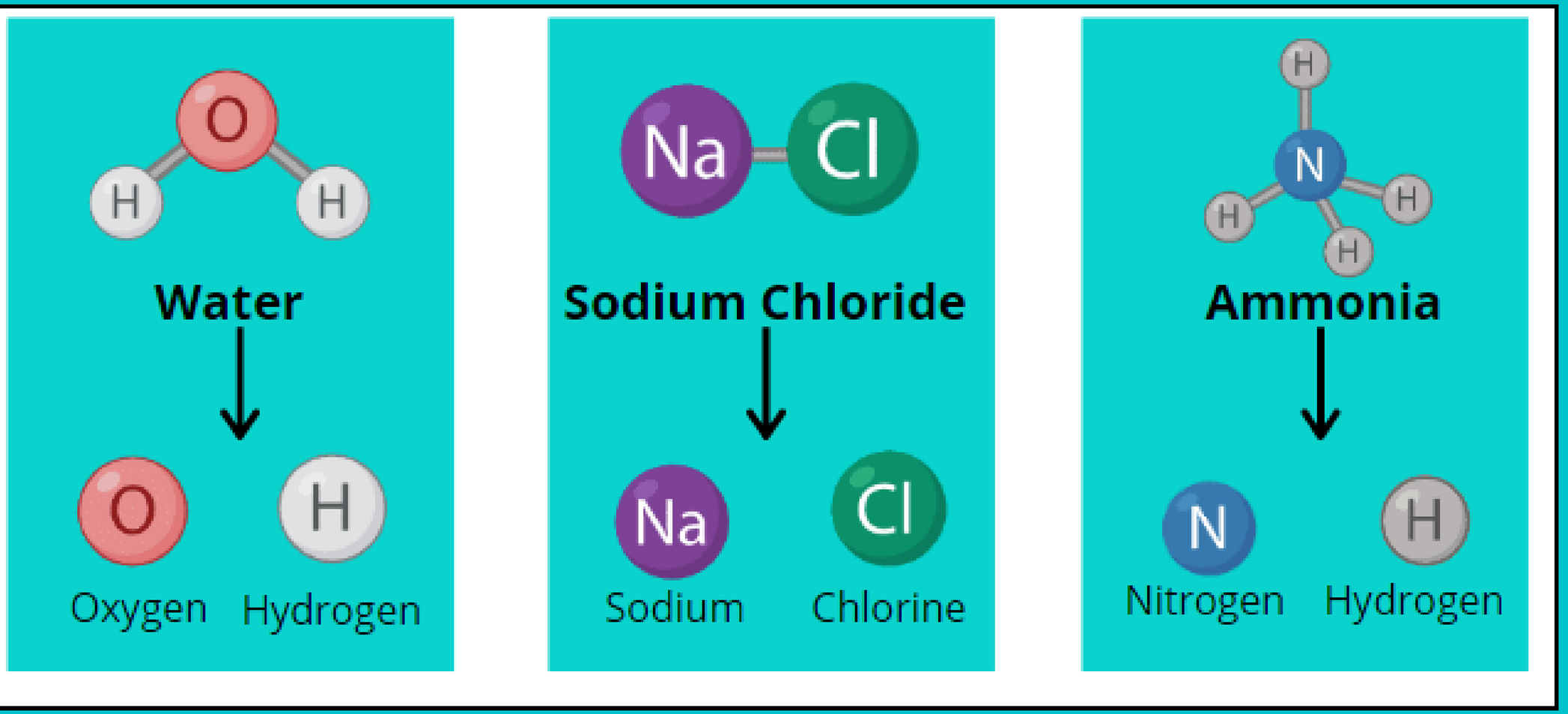
CLASSIFICATION OF MATTER

TYPES OF PURE SUBSTANCE

- Compounds - is a pure substance made of two or more different elements that are chemically joined together.

CLASSIFICATION OF MATTER

COMPOUNDS (EXAMPLES)



CLASSIFICATION OF MATTER

IMPURE SUBSTANCE (MIXTURE)

Is a material that contains two or more different substances mixed together, not chemically combined.

CLASSIFICATION OF MATTER

IMPURE SUBSTANCE (EXAMPLES)



saltwater



air



soil

CLASSIFICATION OF MATTER

HOMOGENOUS MIXTURE

is a mixture where the components are evenly mixed, and you cannot see the different parts.

CLASSIFICATION OF MATTER

HOMOGENOUS MIXTURE

List 5 examples of Homogenous Mixtures

CLASSIFICATION OF MATTER

HETEROGENOUS MIXTURE

is a mixture where the different parts are not evenly mixed, and you can see or separate the individual substances.

CLASSIFICATION OF MATTER

HOMOGENOUS MIXTURE

List 5 examples of Heterogenous Mixtures

CLASSIFICATION OF MATTER

Pure Substance	Impure Substance
• **Only one substance is present**. Eg. Hydrogen (H), Oxygen(O2), water(H2O) • **Constant and definite composition**. Eg Any water molecule is always made up of two hydrogen atoms • **Fixed set of properties**. Eg. Water is always boils at 100 C. • **Can only be separated by chemical methods**. Eg Water (H2O) broken down in to Hydrogen(H) and Oxygen(O2) • **Types of Pure Substance –** ❖ **Elements** ❖ **Compounds**	• **Also known as mixtures.** • **It is a combination of 2 or more substances which does chemically react with each other.**eg Air – nitrogen(N), oxygen(O2), hydrogen(H), carbon dioxide(CO2) and other gases. • **The composition varies**. Eg Water with Impurities / Water with Chlorine. • **Properties vary with the composition**. Eg. Salted water has higher boiling point.